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MicroRNA-1 and microRNA-133a expression are decreased during skeletal muscle hypertrophy

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McCarthy JJ, Esser KA. MicroRNA-1 and microRNA-133a expression are decreased during skeletal muscle hypertrophy. *J Appl Physiol* 102: 306–313, 2007. First published September 28, 2006; doi:10.1152/japplphysiol.00932.2006.—MicroRNAs (miRNAs) are a class of highly conserved, noncoding RNAs involved in posttranscriptional gene regulation. A small number of muscle-specific miRNAs have been identified and shown to have a role in myoblast proliferation and differentiation as well as embryonic muscle growth. The primary objective of the present study was to determine the expression level of the muscle-specific miRNAs in the soleus and plantaris muscles and whether their expression in the plantaris was altered in response to functional overload. Of the miRNAs examined, only miRNA-206 was differentially expressed between soleus and plantaris muscles, as reflected by the sevenfold higher expression in the soleus for both the primary miRNA (pri-miRNA) and mature miRNA (miR). Following 7 days of functional overload, transcript levels for both pri-miRNA-1-2 and pri-miRNA-133a-2 increased by ~2-fold, whereas pri-miRNA-206 levels were elevated 18.3-fold. In contrast, expression of miR-1 and miR-133a were downregulated by ~50% following overload. The discrepancy between pri-miRNA and miR expression following overload was not explained by a change in the expression of components of the miRNA biogenesis pathway, since Drosha and Exportin-5 transcript levels were significantly increased by 50% in response to functional overload, whereas Dicer expression remained unchanged. These results are the first to report alterations in expression of muscle-specific miRNAs in adult skeletal muscle and suggest miRNAs may have a role in the adaptation to functional overload.

gene regulation; skeletal muscle hypertrophy

FIRST DESCRIBED IN THE NEMATODE *Caenorhabditis elegans* as small temporal RNAs required for proper developmental timing, microRNAs (miRNAs) are now recognized as a large class of evolutionarily conserved small RNAs that act to repress gene expression at the posttranscriptional level (24, 30). During the last 2 years, the prevalence of miRNAs in biology has rapidly emerged as reflected in the enormous growth of the miRBase Registry from 506 entries representing 6 organisms to over 3,500 entries from 36 species (16). Remarkably, the latest estimate indicates miRNAs may regulate up to one-third of the mammalian genome, suggesting miRNAs have a central role in regulating gene expression (28).

miRNAs are initially synthesized as a primary transcript (pri-miRNA) with the characteristic 5′ m^7G cap structure and 3′ poly(A) tail of RNA polymerase II transcripts (8, 27). Genomic mapping has revealed pri-miRNAs are often derived from the intron of either protein-coding or noncoding RNA (ncRNA) transcripts and less frequently from the exon of ncRNAs (34). In animals, the mature miRNA (miR) is produced as the result of two endonuclease reactions (15, 26). The first processing step occurs in the nucleus and is carried out by Drosha, a ribonuclease III (RNase III) endonuclease that cleaves the pri-miRNA to release an ~70-bp stem-loop precursor miRNA (pre-miRNA) (25). The pre-miRNA is subsequently transported from the nucleus to the cytoplasm by Exportin-5 where a second RNase III endonuclease, Dicer, processes the stem-loop through a series of cuts to produce an ~21-bp RNA duplex (4, 7, 21, 29, 35). One strand of the duplex, the miR, enters the RNA-induced silencing complex (RISC) and directs the complex to target mRNA (19).

miRNAs are a new class of *trans*-factors that regulate gene expression, which until now have been the exclusive domain of proteins. miRNAs repress gene expression by binding to the 3′-untranslated region (UTR) of target mRNAs and either inhibit translation or promote cleavage of the transcript. Unlike small interfering RNAs (siRNA), miRNA:mRNA interaction requires only the 5′ region (approximately nucleotides 2–8) of the miRNA be complementary to the target mRNA, with the 3′ region of the miRNA apparently serving a modulatory role (13). The mechanism of action, however, does appear to depend on the degree of complementarity between the miRNA and the target; partial complementarity leads to translation inhibition, whereas perfect complementarity results in target mRNA degradation. Given that the "seed" region, nucleotides 2–8, of the miRNA is so short, each miRNA is predicted to target on average 200 genes (28). Despite the fact that only a limited number of target genes have been experimentally confirmed, miRNAs have been shown to function in a range of biological processes, including fat metabolism, glucose homeostasis, cell fate specification, proliferation, and oncogenesis (9, 11, 14, 20, 32, 45).

Bioinformatic analysis of microarray data has identified miRNAs that are highly expressed in a tissue-specific manner in liver, brain, pituitary, pancreas, testis, and striated muscle (38). Numerous miRNA expression profiling studies have consistently shown miRNA-1 (miR-1), miR-133a, and miR-206 to be muscle-specific (3, 10, 36). Of these, miR-1 has been the most extensively studied and found to be one of the most ancient and highly conserved miRNAs (18). Work in *Drosophila* has demonstrated that miR-1 is important for both cardiogenesis, through the regulation of Notch signaling, and skeletal muscle growth during embryonic development (23, 37). Analysis of the presumptive miR-1 promoter has shown that miR-1 expression is regulated by SRF, MyoD, MEF2, and Twist, all factors known to be important in conferring muscle-specific

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expression (5, 45). Chromatin immunoprecipitation experiments have provided additional evidence the myogenic regulatory factors regulate muscle-specific miRNA expression by showing MyoD and myogenin bind the presumptive promoters of these miRNAs (33). Studies using the mouse myogenic C2C12 cell line demonstrated miR-1 and miR-133 are involved in myoblast proliferation and differentiation by regulating the expression of HDAC4 and SRF, respectively (11). A recent study showed miR-206 promotes myogenesis by repressing, in part, the expression of a subunit of DNA polymerase α, the polymerase responsible for DNA synthesis during cell proliferation (22). Genome mapping has revealed the rat noncoding RNA 7H4, identified by Velleca et al. (40), encodes miR-206. Interestingly, 7H4 expression is muscle specific, synaptically enriched, and upregulated on denervation.

Given the importance of the muscle-specific miRNAs in muscle development, it was of interest to determine what role they may have in skeletal muscle plasticity in the adult animal. As an initial effort toward addressing this broad question, the focus of the current study was to determine whether overload-induced hypertrophy altered expression levels of the previously identified muscle-specific miRNAs. The altered expression of miR-1 and miR-133a suggest these muscle-specific miRNAs may have a role in regulating the initial response of skeletal muscle to functional overload.

METHODS

Animal care. The experimental protocol was approved by the University of Kentucky Institutional Animal Care and Use Committee. Male C57BL/6J mice (The Jackson Laboratory), 10 wk of age, were housed in a temperature- and humidity-controlled room and maintained on a 12-h light-dark cycle with food and water ad libitum.

In vivo model of muscle hypertrophy. Mice were randomly assigned to either control or functional overload (FO) group. The bilateral synergist ablation model was used to induce hypertrophy of the plantaris muscle by functional overload as described by Tsika et al. (39). Mice were anesthetized with an intraperitoneal injection of ketamine (100 mg/kg) and xylazine (10 mg/kg). Once anesthetized, a longitudinal incision on the dorsal aspect of the lower hindlimb was made exposing the gastrocnemius muscle. The tendons of the gastrocnemius and soleus muscles were isolated and used to guide in the excision of these muscles without disturbing the plantaris muscle. The incision was closed using a 6-0 silk suture. At the designated time, mice were anesthetized with pentobarbital sodium (50 mg/kg), the plantaris muscles excised, and the animal euthanized by intracardiac injection of saturated KCl. On removal, each muscle was weighed, quickly frozen in liquid nitrogen, and stored at −80°C.

RNA isolation. Total RNA was isolated from plantaris muscle using TRIzol (Invitrogen, Carlsbad, CA) according to manufacturer's directions. RNA samples were treated with TURBO DNase (Ambion, Austin, TX) to remove genomic DNA contamination and RNA integrity assessed using the Agilent 2100 Bioanalyzer (Agilent Technologies, Palo Alto, CA); the average RIN (RNA integrity number) value for all samples was 8.8 ± 0.4 (scale 1–10), indicating high-quality RNA with minimal degradation products. PCR reactions, without the RT step, were performed on each sample using *Rpl*26 primer set to confirm that the DNase treatment effectively removed genomic DNA contamination (data not shown).

RT-PCR analysis. Semi-quantitative PCR was used to detect Drosha, Exportin-5, and Dicer1 transcript levels. First-strand cDNA synthesis from total RNA was performed with oligo$(dT)_{12–18}$ primer using SuperScript II RT (Invitrogen) according to manufacturer's directions. One microliter of the RT reaction was used for end-point PCR analysis with PCR primers designed using Biology Workbench 3.2 PrimerTm program (http://seqtool.sdsc.edu/CGI/BW.cgi). All primers were designed with T_m of ~60°C that amplified 200 bp of the 3′ end of the coding sequence. Primer sequences for each gene are as follows: Drosha, forward 5′-GGATAGGCTGTGGGAAAGGA-3′, reverse 5′-CTTCTTGATGTCTTCAG CCTCC-3′; Exportin-5, forward 5′-CCACTTCAAACGTCTAATCGCT-3′, reverse 5′-GCCG-GAGAAGGAT GCC-3′; Dicer1, forward 5′-TGCTCGAGATG-GAACCAGA-3′, reverse 5′-TCAGCTGTTAGGAACCTGA GGC-3′. To account for any difference in the amount of starting RNA, *Rpl*26 (ribosomal protein L26; forward 5′-CGAGTCCAGC-GAGAGAAGG-3′, reverse 5′-GCAGTCTTTAATGAAAGCC GTG-3′) was chosen as our endogenous control to normalize gene expression because *Rpl*26 expression did not change with FO over the time period examined (control, 5.52 ± 0.36 AU; FO-7, 5.44 ± 0.56 AU).

Detection of primary miRNA transcript. Semi-quantitative PCR was employed, as described above, to detect pri-miRNA transcript levels for pri-miRNA-1-1, -1-2, -133a-1, -133a-2, and -206. miRNA-specific primers were designed to target sequences 170 bp 5′ and 3′ to the miRNA stem-loop (pre-miRNA) as described by Cai et al. (8). Primer sequences for each pri-miRNA with product size in parentheses are pri-miRNA-1-1, forward 5′-ATGAAAA-GGGTTTTGAGACTTTTCA-3′, reverse 5′-GCAAAGTGGCA-GAACA ATG-3′ (395 bp); pri-miRNA-1-2, forward 5′-GGCAT-TGATGGGATCAGGT-3′, reverse 5′-GACTTATCTCTTCAGTAC AGTATAAGGGATG-3′ (400 bp); pri-miRNA-133a-1, forward 5′-GCACTGATGTGAG CTGCAAG-3′, reverse 5′-TTCATGAA-GCTTTTAAGAAACATCTT-3′ (401 bp); pri-miRNA-133a-2, forward 5′-CCATTTT GGGGCA CATAGAG-3′, reverse 5′-TCAGCT-TCCTCCTCTAC TTGCC-3′ (444 bp); pri-miRNA-206, forward 5′-CCCAAC AAGCTCTGCCTG-3′, reverse 5′-GGGAGCATAGT-TGACCTGAAA C-3′ (401 bp). The UCSC Genome Browser in silico PCR tool (http://genome.ucsc.edu) and PCR using mouse genomic DNA (data not shown) were used to verify that each primer set amplified only a single product of the predicted size. Pri-miRNA expression was normalized to *Rpl26* expression.

Detection of mature miRNAs. To detect miR in total RNA samples, the *mir*Vana qRT-PCR miRNA detection kit and *mir*Vana PCR primer sets for miR-1, -24, -133a, and -206 were used according to the manufacturer's directions (Ambion, Austin, TX). Briefly, reverse transcriptase (RT) reactions were performed with microRNA-specific RT primer and 25 ng of total RNA for 30 min at 37°C followed by 10 min incubation at 95°C to inactivate the RT enzyme. End-point PCR was then performed using the RT product and microRNA-specific PCR primer for 20 cycles (two steps: 95°C for 15 s followed by 60°C for 30 s). To account for possible differences in the amount of starting RNA, all samples were normalized to miR-24. We found miR-24 expression did not change with FO (control, 3.02 ± 0.22 AU; FO-7, 2.98 ± 0.58 AU) and so adopted miR-24 as our endogenous control to normalize miR expression. However, miR expression of the soleus and plantaris muscles were normalized to *Rnu6* (U6 small nuclear RNA) levels because miR-24 expression was found to be different between the two muscles. The *Rnu*6 primer set was purchased from Ambion (Austin, TX).

PCR product quantification. Polyacrylamide gel electrophoresis (PAGE) was used to quantify PCR products. The PCR reaction (7.5 μl of 30-μl volume) was loaded onto a 5% gel and run for 1 h at 40 V at room temperature with 0.5× TBE running buffer. The gel was then stained for 30 min at room temperature in SYBR Green I (Invitrogen) according to manufacturer's directions. After staining, gels were imaged using Storm 860 (GE Healthcare, Piscataway, NJ), and bands representing PCR product were quantified using ImageQuant.

Statistical analysis. Data are reported as means ± SE with $n = 5$ or 6 muscles. Student's *t*-test was used to determine whether a significant difference existed between soleus and plantaris muscles or control and FO-7 groups with $P < 0.05$ denoted by an asterisk.

RESULTS

Functional overload induces skeletal muscle hypertrophy. The synergist ablation model was used to induce skeletal muscle hypertrophy by placing a functional overload on the plantaris muscle. Mouse plantaris muscle subjected to 7 days of functional overload significantly increased both muscle wet weight by 45% ($P = 0.002$) and total RNA by 2.7-fold ($P = 0.0004$), confirming the model successfully induced skeletal muscle hypertrophy of the plantaris muscle.

Expression of miRNA-206 is higher in the soleus than in the plantaris. The mature form of miR-1 can be derived from two different pri-miRNAs originating from distinct genomic loci and are termed pri-miRNA-1-1 and pri-miRNA1-2; in a similar fashion, miR-133a can also be generated from two unique loci producing pri-miRNA-133a-1 and pri-miRNA-133a-2. PCR analysis, employing miRNA-specific primers located 170 bp upstream and downstream of the precursor miRNA (pre-miRNA) stem-loop, were used to determine the expression level of each pri-miRNA. Before determining the effect of functional overload on the expression of the muscle-specific miRNAs, we first needed to demonstrate expression of muscle-specific miRNAs in control plantaris muscle. For comparison, we also determined the expression level of miRNAs in control soleus muscle. Expression of each pri-miRNA was detected in control plantaris as well as control soleus muscles (Fig. 1, *A* and *B*). There was no difference between muscles in the level of expression for each of the pri-miRNAs examined except for pri-miRNA-206, which was 7.2-fold higher in the soleus relative to the plantaris (Fig. 1, *A* and *B*). Expression of the corresponding miR paralleled what was observed with the pri-miRNAs since there was no difference in miR-1 and miR-133a expression between soleus and plantaris muscles but a sevenfold greater expression of miR-206 in the soleus compared with the plantaris (Fig. 2, *A* and *B*).

Muscle-specific primary miRNA expression is increased by functional overload. Having established the expression of both the primary and mature muscle-specific miRNAs in the plantaris muscle, we next wanted to determine whether their level of expression was altered after 7 days of functional overload. As shown in Fig. 3, expression of pri-miRNA-1-2 and pri-miRNA-133a-2 increased by ~2-fold in response to overload, whereas pri-miRNA-206 expression increased by ~18.3-fold. In contrast to the other pri-miRNAs, pri-miRNA-1-1 and pri-miR-133a-1 expression did not change following functional overload.

Muscle-specifc miR expression is decreased by functional overload. The observed change in pri-miRNA expression suggested there would be a similar change in the expression of the mature muscle-specific miRNAs miR-1, miR-133a, and miR-206. Surprisingly, PCR analysis revealed overload was associated with a 50% decrease in miR-1 and miR-133a expression levels following overload, whereas miR-206 expression was unchanged (Fig. 4).

Functional overload increased Drosha and Exportin-5 transcript levels. The discordant expression of the pri-miRNAs and the corresponding miR indicate the miRNA biogenesis pathway may be subject to some form of regulation during functional overload. There are three proteins that have been identified as having a major role in miRNA biogenesis; the two RNase III endonuclease Drosha and Dicer and the transport protein Exportin-5. Semi-quantitative PCR analysis determined Drosha and Exportin-5 transcript levels were both increased by 50% following 7 days of functional overload, whereas, in contrast, Dicer transcript levels were unchanged (Fig. 5).

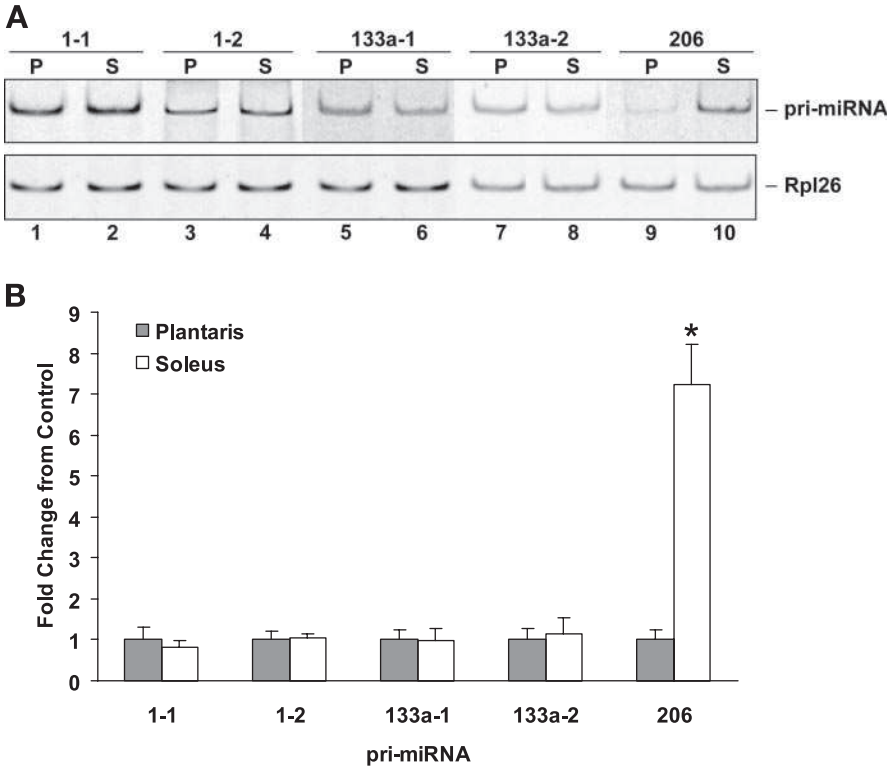

Fig. 1. Expression of muscle-specific primary miRNAs in the mouse plantaris and soleus muscles. *A*: representative image of PCR product from plantaris (P; *lanes 1*, *3*, *5*, *7*, and *9*) and soleus (S; *lanes 2*, *4*, *6*, *8*, and *10*) muscles for pri-miRNA-1-1, -1-2, -133a-1, -133a-2, and -206, respectively. Expression of each pri-miRNA was normalized to Rpl26 expression. *B*: histogram of densitometric quantification of PCR product for each of the pri-miRNAs. There was no difference in the expression level between plantaris and soleus muscle for each of the pri-miRNAs examined except for pri-miRNA-206, which was 7.2-fold higher in the soleus relative to the plantaris ($P = 0.002$). Values are means ± SE ($n = 5$) expressed as fold difference relative to plantaris muscle expression level. *Significant difference ($P < 0.05$).

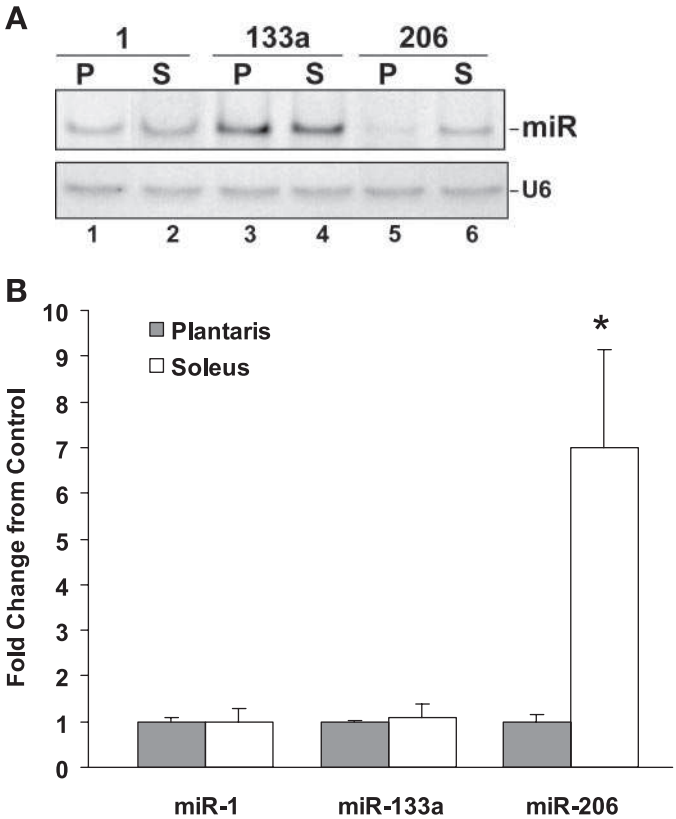

Fig. 2. Expression of muscle-specific miRs in the mouse plantaris and soleus muscles. *A*: representative image of PCR product for miR-1, miR-133a, and miR-206 from plantaris (P; *lanes 1*, *3*, and *5*) and soleus (S; *lanes 2*, *4*, and *6*) muscles, respectively. Muscle-specific miR expression was normalized to U6 expression since it was not different between plantaris and soleus muscles. *B*: histogram of densitometric quantification of respective miR PCR products revealed both miR-1 and miR-133a were expressed at similar levels in the soleus and plantaris muscles in contrast to miR-206, which was found to be significantly elevated sevenfold in the soleus relative to the plantaris (P = 0.0003). Values are means ± SE (n = 5) expressed as fold difference from plantaris expression level. *Significant difference ($P < 0.05$).

DISCUSSION

MiRNAs represent an emerging class of *trans*-factors that repress gene expression by inhibiting translation of target mRNAs. Their high degree of conservation across the metazoa kingdom portends an important role in the regulation of gene expression (18). The current investigation was undertaken as an initial survey of muscle-specific miRNA expression in response to functional overload with the objective of determining if these novel *trans*-factors have a potential role in adult skeletal muscle hypertrophy. The following conclusions can be drawn from the results of this study: *1*) the higher level of expression of miRNA-206 in the soleus relative to the plantaris, as well as the 18.3-fold induction of pri-miRNA with overload, suggest miRNA-206 may be fiber type I specific; *2*) the loss of a direct relationship in the expression level of pri-miRNAs and their corresponding miR is indicative of additional regulation of miRNA biogenesis during functional overload, which does not appear to involve the canonical pathway; and *3*) the decreased expression of miR-1 and miR-133a may serve to promote adaptation to overload by removing posttranscriptional repression of necessary target genes. These results collectively provide the first evidence suggesting miRNAs function during overload-induced muscle hypertrophy and provide the foundation for future studies directed at elucidating the role of specific miRNAs during muscle growth.

The primary aim of the study was to determine whether the expression level of each muscle-specific miRNA was altered in the mouse plantaris muscle after 7 days of functional overload. The analysis employed semi-quantitative PCR to detect expression of each muscle-specific pri-miRNA and its corresponding miR. Conceptually, this strategy is analogous to the more familiar experimental approach of determining mRNA transcript and corresponding protein levels for a given gene of interest in response to some experimental perturbation. In a completely similar fashion to mRNAs, pri-miRNA expression can be measured using RT-PCR because they are derived from RNA polymerase II transcripts and thus polyadenylated. Furthermore, just as proteins can have isoforms generated from distinct transcripts, so mature miRs can be produced from different pri-miRNAs. The muscle-specific pri-miRNAs examined in this study are listed in Table 1. There are two pri-miRNAs for both miR-1 and miR-133a that can be processed to generate each respective miR. Although located on different chromosomes, both pri-miRNAs for miR-1 and miR-133a (pri-miRNA-1-1 and -1-2 for miR-1; pri-miRNA-133a-1 and -133a-2 for miR-133a) give rise to identical miRs (Table 1).

We detected similar levels of expression in the plantaris and soleus muscles for all of the muscle-specific pri-miRNAs and miRs except for miRNA-206. Expression of both pri-miRNA-206 and miR-206 were barely visible in control plantaris muscle but readily detectable in the soleus. This pattern of expression indicates miRNA-206 may be fiber-type I specific given the only difference between the soleus and plantaris muscles of the mouse with respect to fiber type is the expression of either type I myosin or type IIB myosin, respectively (2). This idea is further supported by the striking finding that pri-miRNA-206 expression was increased 18.3-fold following overload, qualitatively similar to the induction of type I myosin in the plantaris in response to functional overload (41). The functional consequence, however, of increased expression of pri-miRNA-206 following overload is not readily apparent given that miR-206 expression was not elevated at the same time point. To determine whether miR-206 does have a potential role in skeletal muscle hypertrophy, future studies will investigate whether miR-206 expression is elevated at later time points following overload.

The possibility that miR-206 may have a role in the regulation of skeletal muscle hypertrophy is supported by a recent study demonstrating miR-206 contributes to the hypertrophic phenotype of Texel sheep (12). Quantitative trait loci mapping revealed Texel sheep have a single nucleotide polymorphism (SNP) within the 3′-UTR of the myostatin gene that results in the formation of a functional miR-1/miR-206 target site. As a consequence of this SNP, translation of the myostatin mRNA is apparently repressed by miR-206, causing a ~70% loss of myostatin serum levels in Texel sheep. As in the myostatin-null mouse, the loss of myostatin protein leads to a dramatic increase in skeletal muscle mass.

Previous studies, as well as findings from this study (compare Figs. 1 and 2), have demonstrated that there is typically a direct relationship between the expression of a pri-miRNA and the corresponding miR (8, 26). This relationship, however, was not found to be the case in response to overload, as demon-

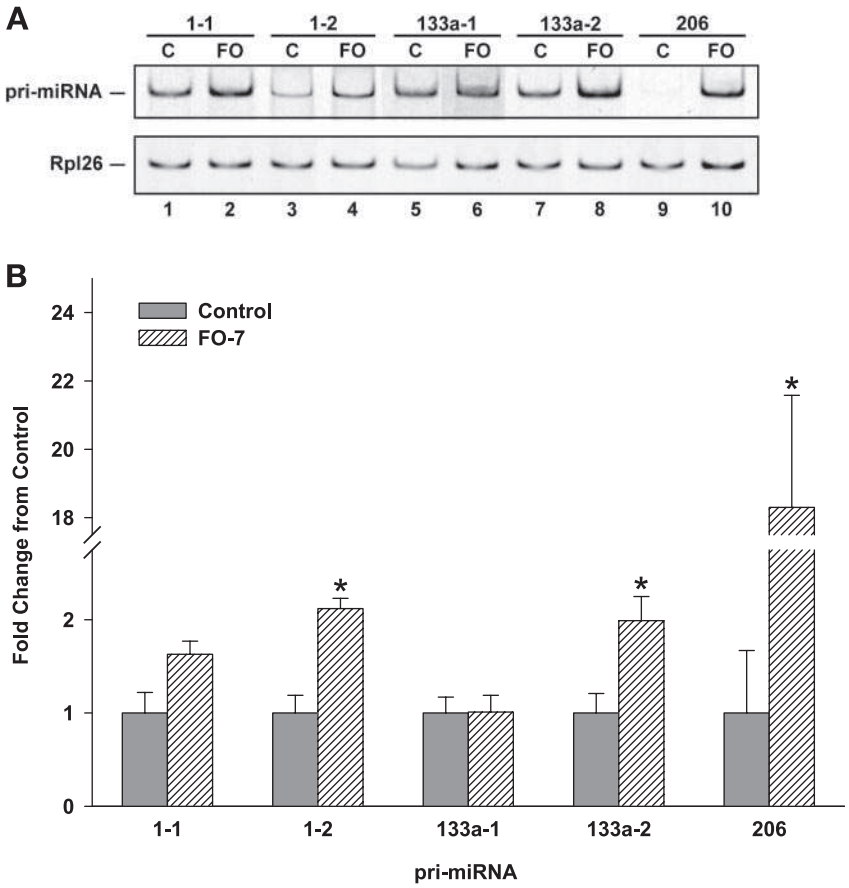

Fig. 3. Expression of primary miRNAs in response to 7 days of functional overload. *A*: representative image of PCR product from control (C; *lanes 1*, *3*, *5*, *7*, and *9*) and functional overload (FO; *lanes 2*, *4*, *6*, *8*, and *10*) groups for pri-miRNA-1-1, -1-2, -133a-1, -133a-2, and -206, respectively. Expression of each pri-miRNA was normalized to Rpl26 expression. *B*: histogram of densitometric quantification of PCR product for each of the pri-miRNAs. Expression of pri-miRNA-1-2 and pri-miRNA-133a-2 were increased ~2-fold ($P = 0.007$ and $P = 0.024$, respectively), whereas pri-miRNA-206 expression was increased by 18.3-fold ($P = 0.014$) in response to overload. Values are means ± SE ($n = 5$) expressed as fold change. *Significant difference ($P < 0.05$).

strated by the approximately twofold increase in pri-miRNA-1-2 and -133a-2 expression, whereas there was a simultaneous 50% decrease in the expression of miR-1 and miR-133a. This discrepancy in expression between pri-miRNA and miR is most notable with miRNA-206; pri-miRNA-206 expression increased by a dramatic 18.3-fold after overload, whereas expression of miR-206 remained unchanged. The loss of coordinated expression between the pri-miRNAs and their downstream product, i.e., miR, suggest with the imposition of functional overload there is an alteration in the normal regulation of miRNA biogenesis.

The pathway involved in the production of a mature miRNA from a pri-miRNA transcript is principally controlled by the activities of three proteins: the two RNase III endonucleases Drosha and Dicer and the transporter Exportin-5. A change in the expression of these components of the miR biogenesis pathway has been shown to alter miRNA levels, as clearly demonstrated in both the mouse and zebrafish Dicer knockouts in which there was a profound loss of miR expression (17, 42). Therefore, a reasonable explanation for discordant expression of pri-miRNA and miR would be that with overload there was a decrease in the expression of one or more of these components of the biogenesis pathway. We found, however, that after 7 days of functional overload of the mouse plantaris muscle, both Drosha and Exportin-5 transcript levels were significantly elevated (Fig. 5), whereas Dicer expression did not change. Unless the protein levels of these components are decreasing, these results indicate an alternative mechanism is responsible for the discrepancy between pri-miRNA and miR level of expression after overload. Two alternative mechanisms that may affect pri-miRNA processing during overload are discussed below.

There is a known RNA editing mechanism that converts adenosine to inosine in double-stranded RNA through a deaminase reaction. The deaminase reaction is carried out by the ADAR (adenosine deaminase, RNA specific) family of enzymes, which were recently shown to regulate the processing of some pri-miRNAs (6, 44). Yang et al. (44) demonstrated editing by ADAR-1 or ADAR-2 of a pri-miRNA suppressed processing by Drosha, which resulted in a loss of the corresponding mature miR. One possible scenario is that, during functional overload, muscle-specific pri-miRNAs are subjected to RNA editing by ADAR, resulting in a decrease in miR expression. This scenario is supported by our microarray analysis of overloaded plantaris muscle, which found ADAR1 transcript elevated by approximately threefold following 7 days of overload (J. J. McCarthy and K. A. Esser, unpublished observation).

An alternative explanation for the discordant expression of pri-miRNAs and miRs with overload is the possible competitive inhibition of Drosha by rRNA. The 2.7-fold increase in total RNA with overload represents a significance increase in the total amount of rRNA given that the vast majority of RNA consists of rRNA. Thus it is possible that during overload pri-miRNAs are not processed by Drosha because the large increase in the amount of rRNA effectively out competes pri-miRNAs for Drosha activity. This possible mechanism is

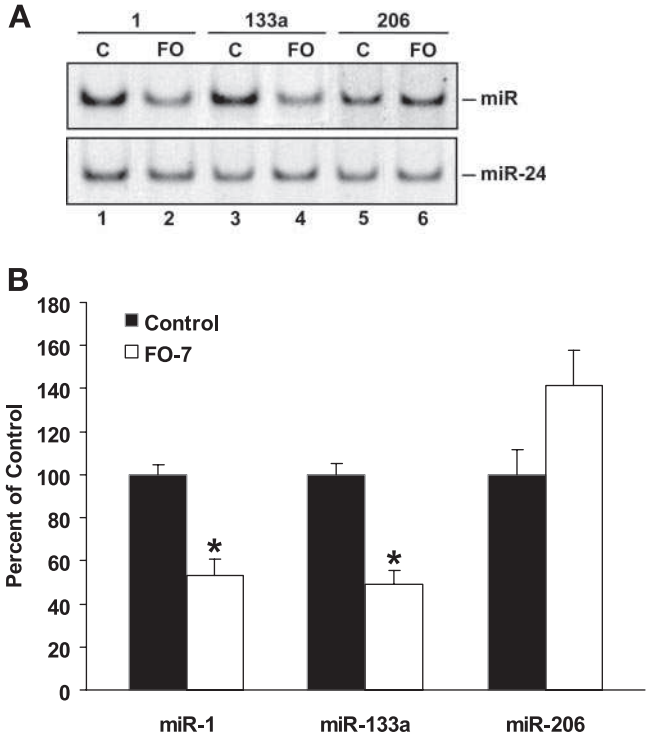

Fig. 4. Expression of muscle-specific miRNAs following 7 days of functional overload. *A*: representative image of PCR product for miR-1, miR-133a, and miR-206 from control (C; *lanes 1*, *3*, and *5*) and functional overload (FO; *lanes 2*, *4*, and *6*) groups, respectively. Muscle-specific miR expression was normalized to miR-24 expression, which was found to not change with functional overload. *B*: histogram of densitometric quantification of respective miR PCR products revealed both miR-1 and miR-133a expression was significantly reduced to ~50% of control ($P = 0.007$ and $P = 0.003$, respectively). Values are means ± SE ($n = 6$) expressed as percent of control. *Significant difference ($P < 0.05$).

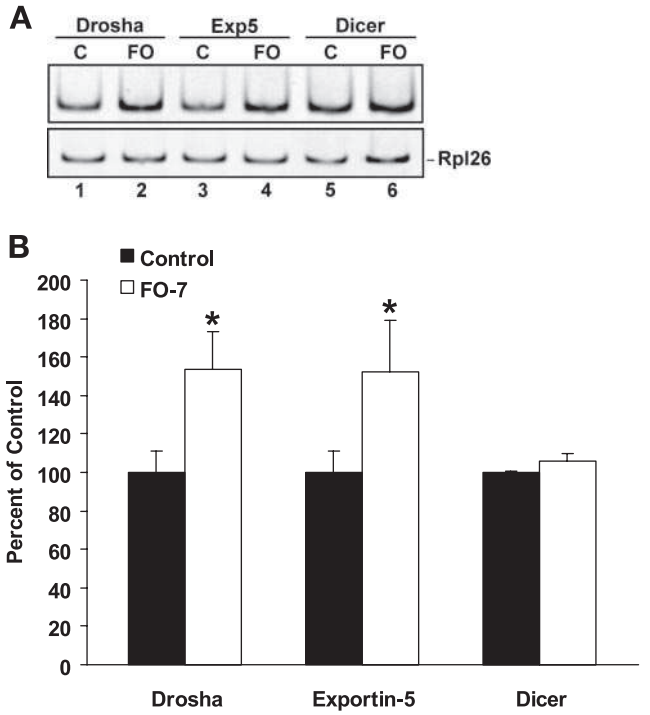

Fig. 5. Transcript levels of components of the miRNA biogenesis pathway following 7 days of functional overload. *A*: representative image of PCR products for Drosha, Exportin-5 (Exp5), and Dicer from control (C; *lanes 1*, *3*, and *5*) functional overload (FO-7; *lanes 2*, *4*, and *6*) groups, respectively. Expression of each transcript was normalized to *Rpl*26 expression. *B*: histogram of densitometric quantification of PCR products showing Drosha ($P = 0.012$) and Exportin-5 ($P = 0.035$) transcript levels were significantly increased by ~50% after 7 days of functional overload, whereas Dicer ($P = 0.251$) transcript levels remained unchanged. Values are means ± SE expressed as percent of control ($n = 5$). *Significant difference from control ($P < 0.05$).

supported by the finding that Drosha has been reported to be involved in pre-ribosomal RNA processing (43).

What is the biological implication for a loss of muscle-specific miR expression during overload-induced hypertrophy? The function of miRs are to repress gene expression by inhibiting translation either by preventing initiation of mRNA translation or promoting mRNA degradation (31). Therefore, a loss of miR expression will lead to increased expression of the protein and/or mRNA of target genes. Predicted target mRNAs of miR-1 and miR-133a suggest their decreased expression with overload may help to promote the expression of genes known to be important for muscle growth, such as c-Met, HGF, IGF-1, SRF, and LIF (Table 1). Of particular interest is the identification of IGF-1 as a potential target of miR-1 because of its well-established role in skeletal muscle hypertrophy. The decreased expression of miR-1 seen at 7 days may, in part, serve to increase IGF-1 protein expression during the initial response to functional overload. In support of this concept, a study by Adams et al. (1) reported IGF-1 peptide levels were signif-

Table 1. *Muscle-specific miRNAs*

Primary miRNA	Mature miRNA	Mouse Chromosome	Host Gene	Predicted Target Genes
1-1	UGGAAUGUAAAGAAGUAUGUA	2 (180106007)	Intergenic	HDAC4*, c-Met,†‡ HGF,†‡ IGF-1†
1-2	UGGAAUGUAAAGAAGUAUGUA	18 (10830826)	Mib 1 intron 12 anti-sense	
133a-1	UUGGUCCCCUUCAACCAGCUGU	18 (10828254)	Mib 1 intron 12 anti-sense	SRF,* Syndecan4,‡ NFATc4,† LIF,‡ HDAC1,‡ HDAC7†
133a-2	UUGGUCCCCUUCAACCAGCUGU	2 (180115338)	Intergenic	
206	UGGAAUGUAAGGAAGUGUGUGG	1 (20900653)	7H4 exon 1	

The "seed" for each miR is underlined. Chromosome location of pre-miR is shown within parentheses. miRNA, microRNA. *Experimentally validated target gene as reported by Chen et al. (11). †and ‡ designate target genes identified from miRBase and miRanda database searches, respectively.

icantly increased in the rat plantaris muscle 12 h after overload, whereas IGF-1 mRNA levels were not significantly elevated until 48 h (1). These results strongly suggest that IGF-1 is post-transcriptionally regulated during the initial stage of overload, and are consistent with a model in which IGF-1 mRNA may be under miR-1 regulation. Although this model is speculative, future studies designed to determine whether IGF-1 is a bona fide target of miR-1 in skeletal muscle following overload should be very interesting.

The results of this study are the first evidence indicating miRNAs may have a role in skeletal muscle hypertrophy. The muscle-specific expression of the miR-1 cluster, which consists of miR-1 and miR-133a, together with the fact it is one of the most ancient miRNA clusters identified, suggests the muscle-specific miRs have an important function in regulating gene expression in striated muscle. An additional exciting possibility is the role miR-206 may have in establishing the type I phenotype by repressing the type II phenotype. The challenge for future studies will be to identify relevant target genes of each muscle-specific miR and how they contribute to the regulation of skeletal muscle growth and phenotype.

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